Homework #1

2.1 2.2 2.3 2.9 2.32

2.1. An antenna has a beam solid angle that is equivalent to a *trapezoidal* patch (*patch with 4 sides, 2 of which are parallel to each other*) on the surface of a sphere of radius r. The angular space of the patch on the surface of the sphere extends between $\pi/6 \le \theta \le \pi/3 (30° \le \theta \le 60°)$ in latitude and $\pi/4 \le \phi \le \pi/3 (45° \le \phi \le 60°)$ in longitude. Find the following:

(a) Equivalent *beam solid angle* [which is equal to number of *square radians/steradians* **or** $(degrees)^2$] of the patch [*in square radians/steradians* ***and*** *in* $(degrees)^2$].

- Exact.
- Approximate using $\Omega_A = \Delta\Theta \cdot \Delta\Phi = (\theta_2 - \theta_1) \cdot (\phi_2 - \phi_1)$. Compare with the exact.

(b) Corresponding antenna *maximum directivities* of part *a* (*dimensionless* ***and*** *in dB*).

解 (a) Exact. $d\Omega = \frac{dA}{r^2} = \sin\theta d\theta d\phi$

故 $\Omega_A = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin\theta d\theta d\phi$

$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} d\phi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin\theta d\theta$

$= \frac{\pi}{12}\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right) = 0.096\ sr = 315.15\ (deg)^2$

Approximate $\Omega_A = \Delta\theta \cdot \Delta\Phi = \frac{\pi}{12} \cdot \frac{\pi}{6} = 0.137 sr = 449.74\ (deg)^2$

相对误差 $\frac{\Delta\Omega_A}{\Omega_A} = 42.7\%$，较大

(b) Exact. $D_0 = \frac{4\pi}{\Omega_A} = 131.14 = 21.18 dB$

Approximate. $D_0 = \frac{4\pi}{\Omega_A} = 91.67 = 19.62 dB$

2.2. Derive (2-7) given the definitions of (2-5) and (2-6)

$$\mathscr{E}(x, y, z; t) = \mathrm{Re}[\mathbf{E}(x, y, z)e^{j\omega t}] \tag{2-5}$$

$$\mathscr{H}(x, y, z; t) = \mathrm{Re}[\mathbf{H}(x, y, z)e^{j\omega t}] \tag{2-6}$$

Using the definitions of (2-5) and (2-6) and the identity $\mathrm{Re}[\mathbf{E}e^{j\omega t}] = \frac{1}{2}[\mathbf{E}e^{j\omega t} + \mathbf{E}^* e^{-j\omega t}]$, (2-3) can be written as

$$\mathscr{W} = \mathscr{E} \times \mathscr{H} = \tfrac{1}{2}\mathrm{Re}[\mathbf{E} \times \mathbf{H}^*] + \tfrac{1}{2}\mathrm{Re}[\mathbf{E} \times \mathbf{H}e^{j2\omega t}] \tag{2-7}$$

解. $\bar{W} = \bar{E} \times \bar{H}$

$= Re[\bar{E}(r) e^{j\omega t}] \times Re[\bar{H}(r) e^{j\omega t}]$

$= \frac{1}{2}[\bar{E}(r) e^{j\omega t} + \bar{E}^*(r) e^{-j\omega t}] \times \frac{1}{2}[\bar{H}(r) e^{j\omega t} + \bar{H}^*(r) e^{-j\omega t}]$

$= \frac{1}{4}[\bar{H}(r) \times \bar{E}^*(r) + \bar{E}(r) \times \bar{H}^*(r) + \bar{E}(r) \times \bar{H}(r) e^{j2\omega t} + \bar{E}^*(r) \times \bar{H}^*(r) e^{-j2\omega t}]$

$\frac{1}{2} Re\{\bar{E}(r) \times \bar{H}^*(r)\} = \frac{1}{4}\{\bar{E}^*(r) \times \bar{H}(r) + \bar{E}(r) \times \bar{H}^*(r)\}$

$\frac{1}{2} Re\{\bar{E}(r) \times \bar{H}(r) e^{j2\omega t}\} = \frac{1}{4}\{\bar{E}(r) \times \bar{H}(r) e^{j2\omega t} + \bar{E}^*(r) \times \bar{H}^*(r) e^{-j2\omega t}\}$

故 $\bar{W} = \bar{E} \times \bar{H} = \frac{1}{2} Re\{\bar{E} \times \bar{H}^*\} + \frac{1}{2} Re\{\bar{E} \times \bar{H} e^{j2\omega t}\}$，证毕

2.3. A hypothetical isotropic antenna is radiating in free-space. At a distance of 100 m from the antenna, the total electric field (E_θ) is measured to be 5 V/m. Find the

(a) power density (W_{rad}) (b) power radiated (P_{rad})

解. $\eta_0 = 377\Omega$

(a) $W_{rad} = \frac{E_0^2}{2\eta_0} = 0.033 W/m^2$，方向指径向外

(b) $P_{rad} = \int W_{rad} d\Omega = W_{rad} 4\pi r^2 = 416.7 W$

2.9. You are an antenna engineer and you are asked to design a high directivity/gain antenna for a space-borne communication system operating at 10 GHz. The specifications of the antenna are such that its pattern consists basically of *one major lobe* and, for simplicity, *no minor lobes* (*if there are any minor lobes they are of such very low intensity and you can assume they are negligible/zero*). Also it is desired that the pattern is symmetrical in the azimuthal plane. In order to meet the desired objectives, the main lobe of the pattern should have a *half-power beamwidth of 10 degrees*. In order to expedite the design, it is assumed that the major lobe of the normalized radiation intensity of the antenna is approximated by

$$U(\theta, \phi) = \cos^n(\theta)$$

and it exists only in the upper hemisphere ($0 \le \theta \le \pi/2, 0 \le \phi \le 2\pi$). Determine the:

(a) Value of n (*not necessarily an integer*) to meet the specifications of the major lobe. *Keep 5 significant figures in your calculations.*

(b) *Exact* maximum directivity of the antenna (*dimensionless* and *in dB*).

(c) *Approximate* maximum directivity of the antenna *based on Kraus'* formula (*dimensionless* and *in dB*).

(d) *Approximate* maximum directivity of the antenna based on *Tai & Pereira's* formula (*dimensionless* and *in dB*).

解: (a) $2\theta_h = HPBW = 10^\circ \Rightarrow \theta_h = 5^\circ$

$$U(\theta)\Big|_{\theta=HPBW} = \cos^n(\theta)\Big|_{\theta=\theta_h}$$

$$\Rightarrow n = \frac{\lg 0.5}{\lg \cos\theta}\Big|_{\theta=\theta_h} = 181.81$$

(b) $U_{max} = 1$,

$$P_{rad} = \int_0^{2\pi}\int_0^{\frac{\pi}{2}} U(\theta,\phi)\sin\theta\, d\theta\, d\phi$$

$$= \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} \cos^n\theta\, d(\cos\theta)$$

$$= 2\pi \frac{1}{n+1}(\cos\theta)^{n+1}\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{2\pi}{n+1}$$

$$D_0 = \frac{4\pi U_{max}}{P_{rad}} = 2(n+1) = 365.62 = 25.63\,dB$$

(c) symmetrical $\Rightarrow HPBW_E^\circ = HPBW_H^\circ = 10^\circ$

Krau's formular:

$$D_0 = \frac{41253}{HPBW_E \times HPBW_H} = 412.53 = 26.15\,dB$$

(d) T&P formular:

$$D_0 = \frac{72815}{HPBW_E^2 + HPBW_H^2} = 364.08 = 25.61\,dB$$

2.32. A base station cellular communication systems *lossless* antenna has a *maximum gain* of 16 dB (above isotropic) at 1,900 MHz. Assuming the *input power* to the antenna is 8 watts, what is the *maximum* radiated power density (in watts/cm^2) at a distance of 100 meters? This will determine the safe level for human exposure to electromagnetic radiation.

解. Lossless Antenna $\Rightarrow e_{cd} = 1$

$G = 16\,dB = 39.81$

$$W(r) = \frac{P_{in} G e_{cd}}{4\pi r^2} = 2.53\times10^{-7}\,w/cm^2$$